Homework 8
Introduction to Complex Analysis
(MATH 4230-40), Spring 2022

Section 3.1 - Convergent Series of Analytic Functions

1. Show that the series $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n^2+z^2}$ converges on the set $\mathbb{C} \setminus \left\{z = ni : n \in \mathbb{Z}\right\}$. Then, show that this convergence is uniform and absolute on each closed disk contained in this region.

To begin, observe that $n^2 + z^2 \neq 0$ when $z \neq \pm in$ for $n = 0, 1, 2, 3, \ldots$. Hence, $\dfrac{1}{n^2+z^2}$ is well-defined when $z \in \mathbb{C} \setminus \left\{z = ni : n \in \mathbb{Z}\right\}$. Taking such a value $z \in \mathbb{C} \setminus \left\{z = ni : n \in \mathbb{Z}\right\}$, let us write $a_n = \left|\dfrac{1}{n^2+z^2}\right|$, $b_n = \dfrac{1}{n^2}$. Observe that

$$\begin{aligned}
\lim_{n\to\infty} \frac{a_n}{b_n} &= \lim_{n\to\infty} \frac{n^2}{|n^2+z^2|} \\
&= \lim_{n\to\infty} \frac{n^2}{n^2|1+z^2/n^2|} \\
&= \lim_{n\to\infty} \frac{1}{|1+z^2/n^2|} \\
&= \frac{1}{\lim\limits_{n\to\infty} |1+z^2/n^2|} \\
&= 1.
\end{aligned}$$

So, by the limit comparison test (the standard calculus 2 result for real series, since $a_n, b_n \in \mathbb{R}$), it follows that $\displaystyle\sum_{n=1}^{\infty} a_n$ converges if and only if $\displaystyle\sum_{n=1}^{\infty} b_n$ converges. Since we know that $\displaystyle\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ does indeed converge (it is a p-series with $p = 2$), it then follows that $\displaystyle\sum_{n=1}^{\infty} a_n$ converges as well. Since $\displaystyle\sum_{n=0}^{\infty}\left|\frac{1}{n^2+z^2}\right| = \frac{1}{|z^2|} + \sum_{n=1}^{\infty} a_n$, the series $\displaystyle\sum_{n=0}^{\infty} \frac{1}{n^2+z^2}$ converges absolutely and, by Proposition 3.1.2, converges point-wise for $z \in \mathbb{C} \setminus \left\{z = ni : n \in \mathbb{Z}\right\}$.

Now, we show that this convergence is uniform and absolute on each closed disk contained in the region $\mathbb{C} \setminus \left\{z = ni : n \in \mathbb{Z}\right\}$. By our previous work above, absolute convergence is given, so all that remains is uniform convergence. To show this, let us consider a disk $\mathrm{D}(z_0, r) \subseteq \mathbb{C} \setminus \left\{z = ni : n \in \mathbb{Z}\right\}$ and observe that $|z| \leq |z_0| + r$ for all $z \in \mathrm{D}(z_0, r)$.

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Hence, if $n > |z_0| + r$, then $n > |z|$ and $n^2 > |z|^2$. So, we have

$$\begin{aligned}
|n^2 + z^2| &= |n^2 - (-z^2)| \\
&\geq \left||n^2| - |z^2|\right| \quad \text{(Reverse Triangle Inequality)} \\
&= \left|n^2 - |z|^2\right| \\
&= n^2 - |z|^2 \quad (\text{Since } n^2 > |z|^2) \\
&\geq n^2 - (|z_0| + r)^2 > 0.
\end{aligned}$$

So, when $n > |z_0| + r$, we have $\left|\dfrac{1}{n^2+z^2}\right| = \dfrac{1}{|n^2+z^2|} \leq \dfrac{1}{n^2 - (|z_0|+r)^2}$. Observe that

$$\sum_{n=1}^{\infty} \frac{1}{n^2+z^2} = \left(\sum_{n \leq |z_0|+r} \frac{1}{n^2+z^2}\right) + \left(\sum_{n > |z_0|+r} \frac{1}{n^2+z^2}\right)$$

and note that the first sum has only a finite number of terms and is therefore finite. In the second series, since $\dfrac{1}{n^2 - (|z_0|+r)^2} \to 0$ as $n \to \infty$, the Weirstrass M-test provides absolute and uniform convergence. Therefore, to our series converges absolutely and uniformly on each closed disk in $\mathrm{C} \setminus \left\{z = ni : n \in \mathbb{Z}\right\}$.

Note: It is important to remember that we are in $\mathbb{C}$, not $\mathbb{R}$! A large number of people claimed that $\left|\dfrac{1}{n^2+z^2}\right| \leq \dfrac{1}{n^2}$, which is not true even for the set $\mathbb{C} \setminus \left\{z = ni : n \in \mathbb{Z}\right\}$, as shown by (for example) the case where $n = 1$ and $z = \dfrac{1}{2}i$, which gives $\left|\dfrac{1}{n^2+z^2}\right| = \dfrac{4}{3} > 1 = \left|\dfrac{1}{n^2}\right|$. Be careful!

2. Show that $\sum_{n=1}^{\infty} \frac{1}{n!z^n}$ is analytic on $\mathbb{C} \setminus \{0\}$. Compute its integral around the unit circle.

First, we show that $\sum_{n=1}^{\infty} \frac{1}{n!z^n}$ is analytic on $\mathbb{C} \setminus \{0\}$. Let $\mathrm{D}(z_0, r) = \{z \in \mathbb{C} : |z_0 - z| \leq r\}$ be any disk contained in $\mathbb{C} \setminus \{0\}$ and note that $|z| > |z_0| - r > 0$ for all $z \in \mathrm{D}(z_0, r)$ (since $\mathrm{D}(z_0, r) \subseteq \mathbb{C} \setminus \{0\}$). On $\mathrm{D}(z_0, r)$, we have $|n!z^n| \geq n!(|z_0| - r)^n > 0$, so

$$\left|\frac{1}{n!z^n}\right| \leq \frac{1}{n!(z_0 - r)^n}.$$

Set $\mathrm{M}_n = \frac{1}{n!(|z_0| - r)^n}$ and recall that $e^z = \sum_{n=0}^{\infty} = \frac{z^n}{n!}$. So, $e^z - 1 = \sum_{n=1}^{\infty} \frac{z^n}{n!}$ and

$$e^{1/(|z_0| - r)} - 1 = \sum_{n=1}^{\infty} \frac{(1/(z_0 - r))^n}{n!} = \sum_{n=1}^{\infty} \frac{1}{n!(z_0 - r)^n}.$$

Therefore, since $\sum_{n=0}^{\infty} \mathrm{M}_n$ converges, the Weirstrass M-test (i.e. Theorem 3.1.7) provides absolute and uniform convergence of $\sum_{n=1}^{\infty} \frac{1}{n!z^n}$ on $\mathrm{D}(z_0, r)$. Since $f_n(z) = \sum_{k=1}^{n} \frac{1}{n!z^k}$ is analytic on $\mathbb{C} \setminus \{0\}$ and converges uniformly on every closed disk in $\mathbb{C} \setminus \{0\}$, it follows that $\sum_{n=1}^{\infty} \frac{1}{n!z^n}$ is analytic on $\mathbb{C} \setminus \{0\}$ by the Analytic Convergence Theorem (i.e. Theorem 3.1.8). Indeed, $\sum_{n=1}^{\infty} \frac{1}{n!z^n} = e^{\frac{1}{z}} - 1$.

Now, we compute the integral of $e^{\frac{1}{z}} - 1 = \sum_{n=1}^{\infty} \frac{1}{n!z^n}$ around the unit circle. By Proposition 3.1.9, we have

$$\begin{aligned}
\int_\gamma \left(\sum_{n=1}^{\infty} \frac{1}{n!z^n} \right) dz &= \sum_{n=1}^{\infty} \left(\int_\gamma \frac{1}{n!z^n} dz \right) \\
&= \sum_{n=1}^{\infty} \frac{1}{n!} \left(\int_\gamma \frac{1}{z^n} dz \right) \\
&= \frac{1}{1!} \int_\gamma \frac{1}{z} dz + \sum_{n=2}^{\infty} \frac{1}{n!} \left(\int_\gamma \frac{1}{z^n} dz \right) \\
&= 2\pi i + \sum_{n=2}^{\infty} \frac{1}{n!}(0) \\
&= 2\pi i.
\end{aligned}$$

3. If $f_n(z) \to f(z)$ uniformly on a region A, and if f_n is analytic on A, is it true that $f_n'(z) \to f'(z)$ on A?

Recall that the Analytic Convergence Theorem states:

Theorem 3.1.8 (Analytic Convergence Theorem) *If* A *is a region in* $\mathbb{C}$ *and* f_n *a sequence of analytic functions on* A *converging to* f *uniformly on every closed disk contained in* A, *then* f *is analytic and* f_n' *converges to* f' *point-wise on* A *and uniformly on every closed disk contained in* A.

Since we have assumed that $f_n(z) \to f(z)$ uniformly on A, it is certainly the case that $f^n \to f$ uniformly on every closed disk contained in A. So, it is indeed true that $f_n' \to f'$ point-wise on A and uniformly on every closed disk contained in A by the Analytic Convergence Theorem. However, while point-wise convergence on A is assured, it need not be the case that $f_n' \to f'$ *uniformly on all of* A–uniform convergence is only assured, in general, on every closed disk contained in A.